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[6]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

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[4]: df = sns.load_dataset("titanic")
df.head()
```

```
[7]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
[7]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column             Non-Null Count  Dtype  
---  --
 0   survived           891 non-null   int64   
 1   pclass             891 non-null   int64   
 2   sex                891 non-null   str      
 3   age               714 non-null   float64  
 4   sibsp             891 non-null   int64   
 5   parch            891 non-null   int64   
 6   fare              891 non-null   float64  
 7   embarked          889 non-null   str      
 8   class             891 non-null   category 
 9   who               891 non-null   str      
10  adult_male        891 non-null   bool     
11  deck              203 non-null   category 
12  embark_town       889 non-null   str      
13  alive             891 non-null   str      
14  alone             891 non-null   bool     
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
memory usage: 80.7 KB
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```
[8]: df.isnull().sum()
```

```
[8]: survived      0
     pclass        0
     sex           0
     age          177
     sibsp         0
     parch         0
     fare          0
     embarked      2
     class         0
     who           0
     adult_male    0
     deck         688
     embark_town   2
     alive         0
     alone         0
     dtype: int64
```

```
[10]: df = df.drop(columns=["deck"])
```

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[11]: df["age"] = df["age"].fillna(df["age"].mean())
     df["embarked"] = df["embarked"].fillna(df["embarked"].mode()[0])
```

```
[12]: df.isnull().sum()
```

```
[12]: survived      0
     pclass        0
     sex           0
     age           0
     sibsp         0
     parch         0
     fare          0
     embarked      0
     class         0
     who           0
     adult_male    0
     embark_town   2
     alive         0
     alone         0
     dtype: int64
```

```
[13]: from sklearn.preprocessing import StandardScaler

     scaler = StandardScaler()

     df[["age", "fare"]] = scaler.fit_transform(df[["age", "fare"]])
```

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[14]: df[["age", "fare"]].head()
```

```
[14]:
```

	age	fare
0	-0.592481	-0.502445
1	0.638789	0.786845
2	-0.284663	-0.488854
3	0.407926	0.420730
4	0.407926	-0.486337

```
[15]: from sklearn.model_selection import train_test_split

X = df[["age", "fare", "sex", "pclass"]]
y = df["survived"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)

Training Data Shape: (712, 4)
Testing Data Shape: (179, 4)
```

```
[16]: df.to_csv("preprocessed_titanic.csv", index=False)

print("Dataset saved successfully!")

Dataset saved successfully!
```

```
[ ]:
```

